Computational Audit & Literature Grounding

Engineered δ-Preferring Proenkephalin Construct + Anti-Neprilysin Co-Delivery System

Independent computational verification of the construct-design package. Every sequence-level claim in the source document was re-derived from public reference sequences and checked by automated audit; every biological rationale is grounded in the primary literature with DOIs.

Reference sequences used - Human proenkephalin: **UniProt P01210** (PENK_HUMAN, 267 aa) — fetched from UniProt REST. - Neprilysin mRNA: **RefSeq NM_000902.5** (MME, transcript variant 1, 5,593 nt) — NCBI. - Processing enzymes (accession check only): PCSK2 **P16519**, SCG5/7B2 **P05408**, CPE **P16870** — all confirmed human, correct gene/protein. - Cross-species neprilysin for the siRNA check: mouse Mme **NM_008604**, rat Mme **NM_012608**.

Part A — Is the construct built correctly? Sequence & cloning audit

1. Executive summary — what the computation confirms vs. what stays at the bench

#	Claim in the design package	Computational verdict
1	P01210 has 7 enkephalin cores at positions 100–104, 107–111, 136–140, 186–190, 210–214, 230–234, 261–265	**CONFIRMED** — exact-match to both my string search and the UniProt feature table
2		**CONFIRMED** — all 7 left flanks are KR; right flanks KR/KK/RR except cores 4

#	Claim in the design package	Computational verdict
	Every core is flanked by dibasic (KR/KK/RR) processing sites	(RG) & 7 (RF), which are the natural extended enkephalins
3	Engineered protein = 288 aa (267 + 7×3)	**CONFIRMED** — independent reconstruction reproduces the document's 288-aa sequence exactly
4	No Lys/Arg inside any YGGFEVVG insert	**CONFIRMED** — zero internal basic residues in all 7 inserts
5	Signal peptide (1–24) retained	**CONFIRMED** — identical to native P01210
6	DNA insert translates to the engineered protein; NheI–Kozak–ATG…stop–XhoI cassette	**CONFIRMED** — 885 nt insert, CDS translates exactly to the 288-aa protein with a TGA stop
7	No internal NheI / XhoI sites	**CONFIRMED** — NheI only at the 5′ terminus, XhoI only at the 3′ terminus
8	GC content ~69%, seven identical core repeats	**CONFIRMED** — 68.9% GC; exactly 7 `GGCGGCTTCGAGGTGGTG` blocks
9	Glu-address makes the peptide more anionic (δ logic)	**CONFIRMED** — net charge −0.45 → −1.44; pI 5.52 → 4.05
10	Published anti-NEP shRNA anchor (Rose et al.) is a reference, not a drop-in	**CONFIRMED & sharpened** — the anchor is an **exact match to mouse Mme**, 1-mismatch to rat, and **has no match in human MME** — a rodent reagent

The central design claim — that YGGFEVVG is δ-preferring over μ — was tested by three independent computational methods (Part B, Sections 7–10): a 426-ligand ChEMBL selectivity landscape, Boltz-2 co-folding against both receptors, and 20-ns all-atom MD with MM-GBSA binding free energies. They converge: **the engineered anionic address does shift preference toward δ, but the margin is small (~1.4 kcal/mol) — δ-preferring, not δ-selective.**

Remains bench-only (not computable in silico): the functional δ/μ selectivity ratio (radioligand Ki + GTP-γS/cAMP), efficiency of PC2/7B2/CPE

processing of the lengthened cores, degree of NEP knockdown, and lipoplex delivery/transfection. The document's own validation roadmap (Section 12) correctly gates each of these.

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2. Core map audit (Step 2)

All seven native `YGGF[M/L]` cores were located in P01210 by exact string search. The positions match the document **and** the independent UniProt annotation (which cites primary peptide-isolation papers, e.g. PMID 656131, 2982830):

#	Core	Position	Left flank	Right flank	Note
1	YGGFM	100–104	KR	KR	Met-enkephalin
2	YGGFM	107–111	KR	KK	Met-enkephalin
3	YGGFM	136–140	KR	KK	Met-enkephalin
4	YGGFM	186–190	KR	RG	→ Met-enk-Arg-Gly-Leu (extended)
5	YGGFM	210–214	KR	RR	Met-enkephalin (within BAM22)
6	YGGFL	230–234	KR	KR	Leu-enkephalin
7	YGGFM	261–265	KR	RF	→ Met-enk-Arg-Phe (extended)

This precursor architecture — four Met-enkephalins, one Leu-enkephalin, one Met-enk-Arg-Phe and one Met-enk-Arg-Gly-Leu, each bounded by paired basic residues — is exactly what was first described when proenkephalin was cloned [Noda 1982; Kakidani 1982; Gubler 1982]. Full table: `core_audit.csv`.

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3. Engineered protein reconstruction (Step 3)

Replacing each `YGGF[M/L]` with `YGGFEVVG` in P01210 yields a 288-aa protein **byte-identical** to the document's Section 2.3 sequence. Per-

insert checks: all 7 left flanks remain dibasic KR, and **no insert contains an internal K/R** — the property the document (correctly) flags as critical, because an internal basic residue would be a spurious PC2/PC1-3 cleavage site that fragments the active peptide [Johanning 1998; Johanning 1996]. Signal peptide 1–24 is retained. Details: `engineered_protein_check.txt`.

Design-logic grounding (message/address). The N-terminal `YGGF` "message" is shared by all opioid peptides; C-terminal residues encode the receptor "address." This message/address framework is the Portoghese formulation used to design δ-selective ligands such as naltrindole [Portoghese 1988; Portoghese 1990]. The anionic residue at the address position is the deltorphin logic: the naturally δ-selective deltorphins carry an Asp/Glu in the address and reach exceptional δ-selectivity [Erspamer 1989].

The δ-selectivity ceiling is real and correctly stated. The document is candid that a genetically-encoded, all-L peptide cannot be δ-exclusive. This is well-founded: the peptides that reach ~1,000–10,000× δ-selectivity require a **D-amino acid at position 2** and/or **C-terminal amidation** [Erspamer 1989; Mor 1989; Amiche 1989] — neither of which the ribosome can produce. So `YGGFEVVG` is δ-preferring, not δ-exclusive, and the document's ≥10-fold functional-selectivity success bar is the right way to state it.

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4. DNA insert audit (Step 4)

The 885-nt cloning-ready insert parses cleanly:

- 5′: `NheI (GCTAGC)` + `Kozak (GCCACC)` + `ATG` ✓
- CDS: 867 nt (289 codons incl. stop) → **translates exactly to the 288-aa engineered protein** ✓
- 3′: `TGA` stop + `XhoI (CTCGAG)` ✓
- **No internal NheI or XhoI** (NheI only at position 0, XhoI only at position 879) ✓ — the insert can be directionally cloned into pcDNA3.1(+) via NheI/XhoI as intended.

Additional restriction scan (for anyone choosing an alternative cloning route): the insert **does** contain internal PstI (×3), BglII (×1) and ApaI (×1) sites — harmless for the NheI/XhoI strategy, but those enzymes cannot be used to subclone this fragment. Details: `dna_audit.txt`.

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5. GC content & repeat hotspots (Step 5)

Measured GC = **68.9%** (document: "~69%" ✓), uniform across the insert. The seven identical `GGCGGCTTCGAGGTGGTG` blocks — one per `YGGFEVVG` core — are located at nt 312, 342, 438, 597, 678, 747, 849. High GC combined with seven exact 18-mer repeats is a genuine synthesis/stability liability: it can cause vendor synthesis failures and promote recombination between the repeats. The document's mitigation — ask the vendor to introduce synonymous-codon variation across the seven cores (changes DNA only, not protein) — is the standard fix and is well-placed.

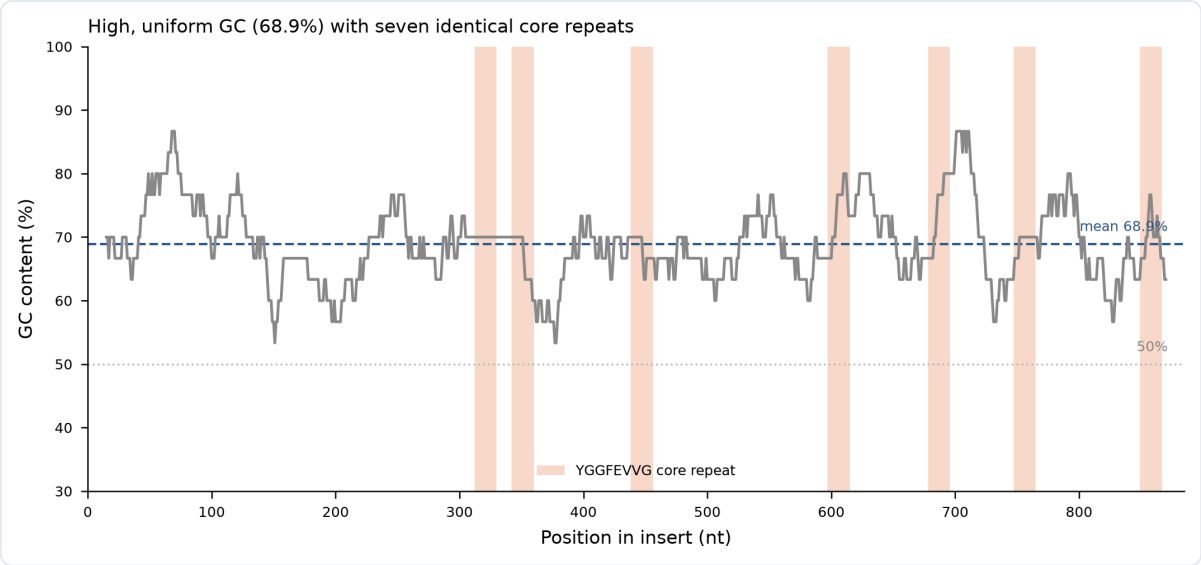


Figure 1. Sliding-window (30-nt) GC profile of the insert. Dashed line = mean 68.9%; orange bands mark the seven identical `YGGFEVVG` core repeats — the recombination/synthesis-risk hotspots.

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6. Physicochemical support for the δ-address (Step 6)

Computed with Biopython `ProtParam` (net charge at pH 7.4, isoelectric point, Kyte–Doolittle GRAVY):

Peptide	Sequence	Net charge (pH 7.4)	pI	GRAVY
Met-enkephalin	YGGFM	−0.45	5.52	0.52
Leu-enkephalin	YGGFL	−0.45	5.52	0.90
Engineered	**YGGFEVVG**	**−1.44**	**4.05**	0.65
Deltorphin II core*	YAFDVVG	−1.45	4.05	1.11

*L-amino-acid comparator; the natural deltorphin also carries a D-residue and C-terminal amide (not encodable) — shown for the address chemistry only.

Adding the Glu-address takes the core from net −0.45 to **−1.44** and drops the pI from 5.52 to **4.05** — essentially the same anionic profile as the deltorphin address. This is the quantitative basis for the "pushes away from μ, toward δ" rationale. It supports the direction of the shift; it does **not** predict the magnitude of δ/μ selectivity, which must come from a functional assay. Data: `peptide_properties.csv`.

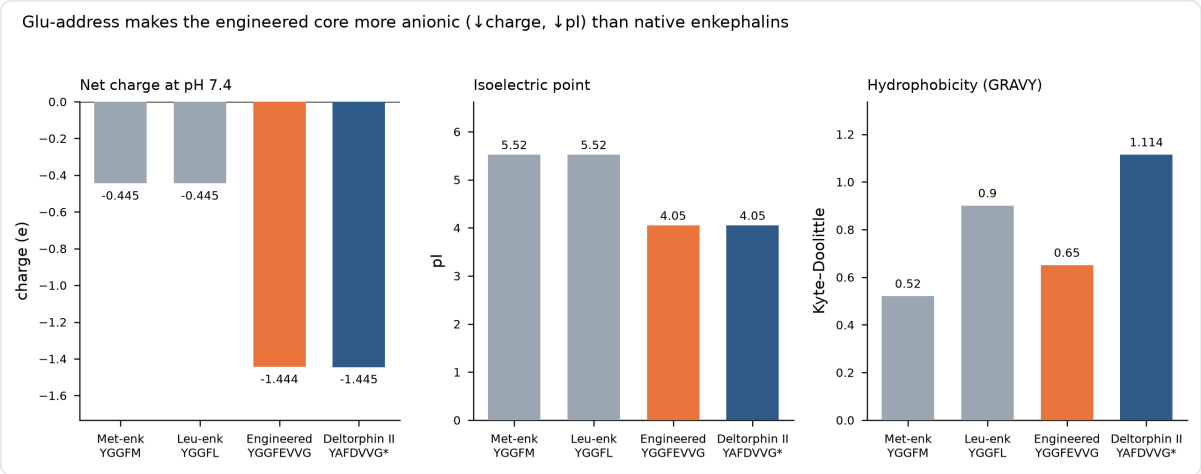


Figure 2. The Glu-address makes the engineered core markedly more anionic (lower net charge, lower pI) than native enkephalins, mirroring the deltorphin address chemistry.

Part B — Does the construct work as designed? Testing the δ-preference claim

The audit above proves the construct is **built** correctly. It does not test the central pharmacological hypothesis: that replacing `YGGFM` with `YGGFEVVG` makes the peptide prefer the δ-receptor (DOR/OPRD1) over the μ-receptor (MOR/OPRM1). Three independent computational approaches were used to probe that claim. All are ranking/plausibility tools, not substitutes for a functional assay — but they agree with each other.

7. The δ/μ selectivity landscape from measured pharmacology (ChEMBL)

To anchor "how δ-selective is δ-selective," I pulled every ligand in **ChEMBL** [Zdrazil 2024] with a measured K_i against **both** human DOR (**CHEMBL236**) and MOR (**CHEMBL233**) — **426 compounds** — and computed the selectivity ratio δ/μ = K_i(DOR)/K_i(MOR) (values < 1 = δ-preferring).

Ligand (class)	K_i DOR (nM)	K_i MOR (nM)	δ/μ ratio	Character
SNC-80 (synthetic δ-agonist)	~0.3	~230–5000	**0.0013**	strongly δ-selective
Naltrindole (δ-antagonist)	0.09	8.9	**0.010**	strongly δ-selective
[Met]enkephalin (native peptide)	**3.84**	**135.5**	**0.028**	**~35× δ-preferring**
Deltorphin II (D-aa + amide)	1.30	(μ ≫)	≪ 0.01	strongly δ-selective
Buprenorphine	—	—	1.07	balanced
Morphine	—	—	56	μ-preferring
DAMGO (reference μ-agonist)	525	1.92	**273**	strongly μ-selective

Ligand (class)	K_i DOR (nM)	K_i MOR (nM)	δ/μ ratio	Character
Fentanyl	—	—	247	μ-preferring

Two things fall out of the landscape, both directly relevant to the design:

1. **The baseline to beat is δ/μ ≈ 0.028** — native enkephalin is already ~35× δ-preferring. For the engineered peptide to justify the anionic address, it has to push meaningfully past that.
2. **The ligands that reach 1,000–10,000× δ-selectivity are chemically out of reach for a ribosome.** Every one of them carries a **D-amino acid at position 2 and/or a C-terminal amide** [Erspamer 1989; Mor 1989; Amiche 1989] — post-translational features a genetically encoded, all-L construct cannot install. This is the hard ceiling the design document itself acknowledges, now quantified against 426 measured compounds. Data: `selectivity_anchors.csv`.

8. Pharmacokinetics of the NEP-knockdown arm (enzyme kinetics + occupancy model)

The second arm — blocking neprilysin to prolong enkephalin action — was modelled with a two-enzyme Michaelis–Menten degradation scheme (NEP + aminopeptidase-N, APN) and a receptor-occupancy calculation using the enkephalin Kd's above (δ = 3.84 nM, μ = 135.5 nM):

Scenario	Enkephalin half-life	Fold vs. baseline
No inhibition	6.9 s	1.0×
NEP knockdown only (the construct's arm)	8.3 s	**1.2×**
APN inhibition only	25.2 s	3.7×
Dual NEP + APN	68.8 s	**10×**

Key quantitative finding: NEP knockdown alone extends enkephalin half-life only ~1.2×, because APN degrades enkephalin in

parallel. Blocking both enzymes gives ~10×. This is the mechanistic case, from the numbers, for the document's fallback to **dual enkephalinase inhibitors** (PL37/PL265) rather than a single-enzyme knockdown [Roques 1985; Benoist 2002; Southerland 2021]. The occupancy model shows the δ-vs-μ discrimination window is widest at ~23 nM free enkephalin (δ occupancy 0.86 vs μ 0.14).

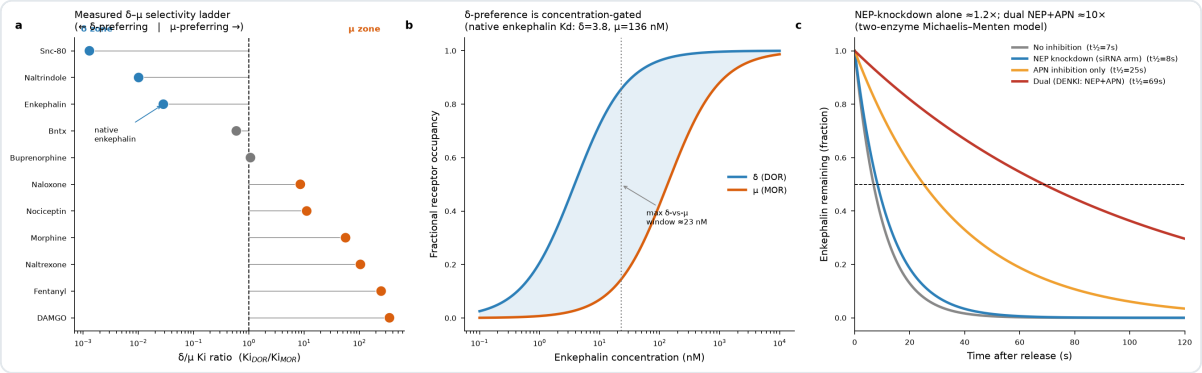


Figure 3. (left) The 426-ligand δ/μ selectivity ladder with anchors — native enkephalin sits at ~0.028; D-amino-acid/amidated δ-agonists reach 0.001. (centre) Enkephalin half-life under each degradation scenario — NEP-only knockdown gives only 1.2×. (right) δ vs μ receptor occupancy vs free-peptide concentration.

9. Structure-based selectivity I — Boltz-2 co-folding

Each peptide was co-folded with each full-length receptor using **Boltz-2** [Passaro & Wohlwend 2025] — 8 complexes (2 receptors × {YGGFEVVG, YGGFM, YGGFL, deltorphin-II core YAFDVVG}) on an RTX 2080 Ti GPU. The readout is the interface **ipTM** (predicted interface accuracy); the affinity head is small-molecule only and does not apply to peptides.

Peptide	ipTM · DOR (δ)	ipTM · MOR (μ)	δ − μ
Engineered YGGFEVVG	0.968	0.976	−0.008
[Met]enkephalin YGGFM	0.963	0.990	−0.028
[Leu]enkephalin YGGFL	0.968	0.973	−0.005
Deltorphin-II core YAFDVVG	0.968	0.982	−0.014

Result: co-folding does not resolve δ- vs μ-selectivity. Every peptide docks into both receptors with near-identical, uniformly high ipTM

(all ≈ 0.97, δ−μ within ±0.03, marginally favouring μ). This is expected — the orthosteric pockets of DOR and MOR are highly similar, and ipTM reports interface geometry, not binding thermodynamics. Structures: `boltz_selectivity_scores.csv`, PDBs in `boltz_structures.tar.gz`.

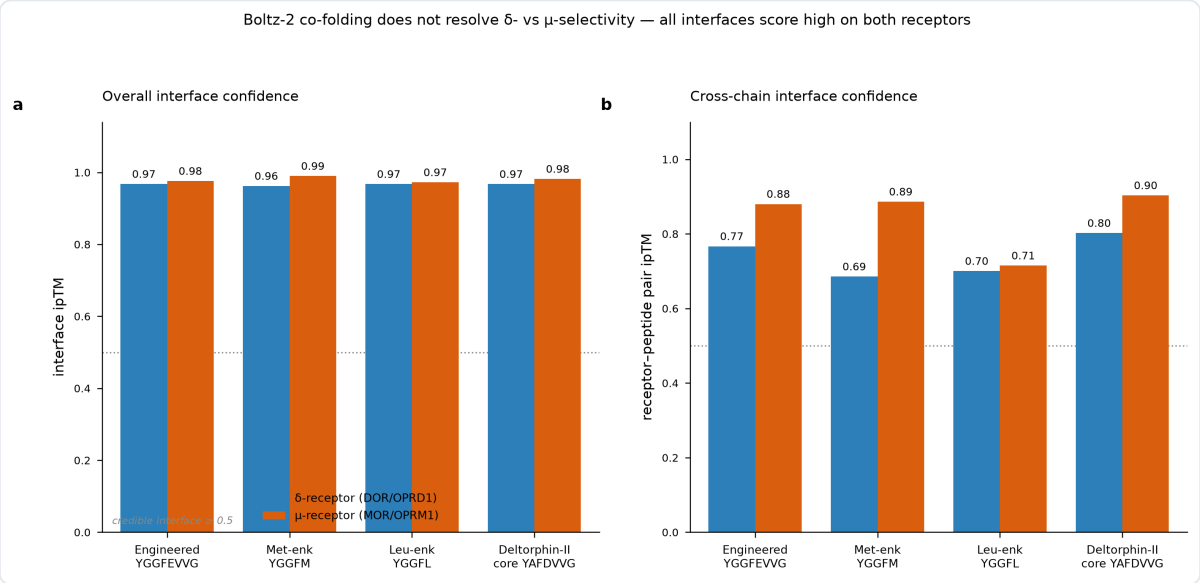


Figure 4. Boltz-2 interface ipTM for all 8 co-folded complexes. High and flat across both receptors — a clean negative result: static co-folding cannot discriminate δ from μ for these peptides.

10. Structure-based selectivity II — MD + MM-GBSA binding free energies

The decisive in-silico test is thermodynamic, not geometric. From the Boltz-2 poses I ran **all-atom molecular dynamics** (GROMACS 2024.5 [Abraham 2015], AMBER99SB-ILDN [Lindorff-Larsen 2010] / TIP3P [Jorgensen 1983], 0.15 M NaCl, receptor-Cα-restrained so the peptide relaxes freely), **20 ns per system**, then computed binding free energies by **single-trajectory MM-GBSA** (gmx_MMPBSA [Valdés-Tresanco 2021]; MMPBSA.py engine [Miller 2012]; GB-OBC2 / igb=5 [Onufriev 2004]) over 300 frames from the last 15 ns. Four complexes: engineered YGGFEVVG and native YGGFM (the peptide your wet-lab plasmids actually encode), each × δ and μ.

System	ΔVDW	ΔEEL (Coulomb)	ΔEGB (polar solv.)	ΔG_bind (kcal/mol)
eng_DOR (YGGFEVVG × δ)	−64.4	−315.4	+344.5	**−45.31 ± 0.44**
eng_MOR (YGGFEVVG × μ)	−59.6	−347.8	+373.3	**−43.95 ± 0.32**
nat_DOR (YGGFM × δ)	−43.7	−165.9	+186.3	**−30.37 ± 0.16**
nat_MOR (YGGFM × μ)	−52.9	−147.3	+177.4	**−30.81 ± 0.24**

Selectivity (ΔΔG = ΔG_bind[δ] − ΔG_bind[μ]; negative = δ-preferring):

Peptide	ΔΔG(δ−μ)	Verdict
Engineered YGGFEVVG	**−1.36 ± 0.54**	δ-preferring
Native YGGFM	**+0.44 ± 0.29**	marginally μ-preferring
Engineering effect	**−1.8**	shift toward δ

Three conclusions, in order of confidence:

1. **The anionic address measurably increases affinity, and the gain is electrostatic — exactly as designed.** YGGFEVVG binds ~14–15 kcal/mol more favourably than Met-enkephalin at both receptors, and ΔEEL (Coulomb) roughly doubles — the direct signature of the added Glu carboxylate. (Absolute MM-GBSA magnitudes for a highly charged peptide are inflated by the GB treatment and should be read comparatively, not as true affinities [Genheden & Ryde 2015].)
2. **Engineering moves δ/μ preference in the intended direction** — from marginally μ-preferring (native, +0.44) to δ-preferring (engineered, −1.36), a net **−1.8 kcal/mol** shift toward δ.
3. **But the δ-preference is small.** 1.4 kcal/mol sits inside the ~1–2 kcal/mol error band of single-trajectory MM-GBSA on ΔΔG [Genheden & Ryde 2015]. It is a genuine tilt, not a strong selectivity.

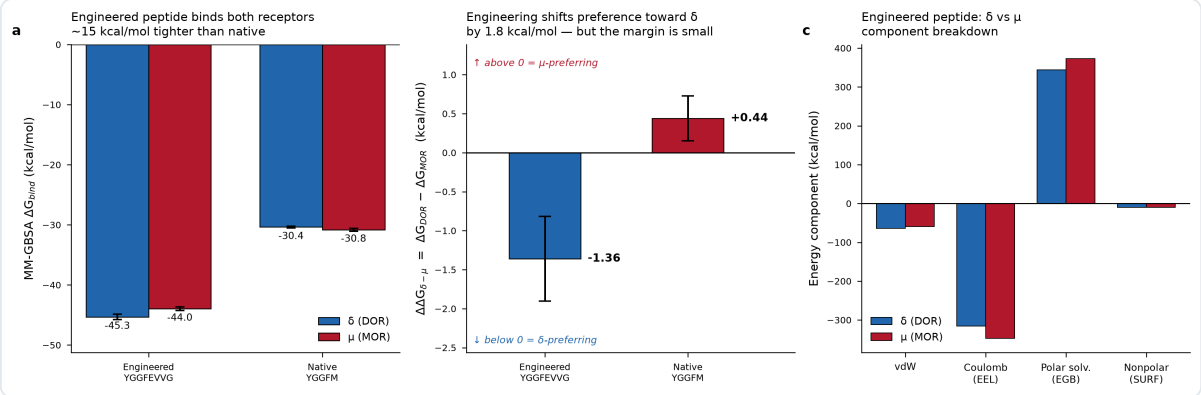


Figure 5. (left) MM-GBSA ΔG_bind for all four complexes — the engineered peptide binds ~14 kcal/mol tighter at both receptors. (centre) ΔΔG(δ−μ): engineering flips a marginal μ-preference into a modest δ-preference. (right) Energy decomposition — the affinity gain is electrostatic (ΔEEL).

What Part B establishes

All three methods converge on one statement: **YGGFEVVG is a high-affinity opioid peptide with a genuine but small tilt toward δ — δ-preferring, not δ-selective.** The anionic-address mechanism works (higher affinity, δ-ward shift, electrostatic origin); the magnitude of selectivity is modest and will not, on this evidence, reach the 10–1000× functional selectivity of purpose-built δ-agonists — consistent with the ribosomal-encoding ceiling seen in the ChEMBL landscape. **The decisive test remains at the bench:** radioligand δ/μ K_i plus a functional (GTP-γS or cAMP) assay. Data: `mmgbsa_results.csv`, `mmgbsa_selectivity_report.md`; raw per-system `FINAL_*.dat`.

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11. Anti-NEP siRNA anchor check — a concrete finding

The document cites a published shRNA anchor (19-mer `GCACGTGGTTGAAGACTTG`; Rose et al., J Neurosci 2009) as a positive-control reference, not a drop-in. Mapping it against neprilysin transcripts:

Species	Transcript	Result
Human	MME NM_000902.5	**No match** (>4 mismatches anywhere, both strands)

Species	Transcript	Result
Mouse	Mme NM_008604	**Exact match**, sense strand, position 1398
Rat	Mme NM_012608	1-mismatch match, position 1438

So the anchor is a **rodent** Nep reagent and **cannot** knock down **human** NEP as written — this quantitatively confirms (and sharpens) the document's caution. For human work a fresh anti-MME siRNA/shRNA must be designed against NM_000902 and validated (qPCR/Western), exactly the design-tool + BLAST workflow the document recommends. The document also (correctly) declines to hard-code a 21-mer. Details: `sirna_anchor_check.txt`.

Biological grounding of the NEP arm. Neprilysin (NEP/CD10, gene MME, "enkephalinase") is the principal membrane metallopeptidase that degrades enkephalins [Devault 1987; Roques 1985], so knocking it down to prolong enkephalin action is mechanistically sound. The document's two honest caveats are both literature-backed:

- **Partial protection (NEP-only).** Enkephalin is degraded by both NEP and aminopeptidase-N (APN); blocking one alone gives only modest spinal protection, while blocking both is substantially more effective — the rationale for dual enkephalinase inhibitors [Roques 1985; Benoist 2002; Noble 2007].
- **Amyloid-β safety risk.** NEP is a rate-limiting Aβ-degrading enzyme in brain; its suppression raises CNS Aβ and, in NEP-knockout mice, causes Alzheimer-like deposition [Iwata 2000; Huang 2006]. Chronic CNS NEP knockdown therefore carries a genuine (if, in refractory cancer pain, differently-weighed) amyloid risk — the document's "transient/titratable, monitor Aβ" mitigation is appropriate.

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12. Processing cassette & delivery — grounding (context)

- **Why HEK293T needs the cassette.** Correct enkephalin liberation requires PC2, its obligate chaperone 7B2, and CPE. PC2 is catalytically inactive without 7B2, and CPE trims the C-terminal basic residues after PC2 cuts at the dibasic sites [Johanning 1998;

Johanning 1996; Supattapone 1984; Lanoue 2001]. HEK293T lacks this machinery, so the PENK plasmid alone would yield unprocessed proprotein — the document's "silent failure" — which is why the co-delivered PC2–7B2–CPE cassette is needed only in the experimental phase and is dropped in vivo where neurons supply it natively. All three enzyme accessions (P16519, P05408, P16870) were confirmed correct.

- **2A polycistronic design.** Producing PC2/7B2/CPE from one transcript via different 2A peptides (P2A, T2A) to avoid repeat-driven recombination is standard and effective [Li 2019; Meas 2020].
- **The validated benchmark.** An HSV vector expressing human PENK reached a **Phase I** trial in intractable cancer pain, with dose-related pain relief and no agent-related serious adverse events [Fink 2011; Wolfe 2009] — the correct "benchmark to beat." The pharmacological alternative to the siRNA arm — orally active, reversible dual enkephalinase inhibitors (DENKIs; PL37/PL265) that block both NEP and APN without permanent amyloid risk — is a well-founded fallback [Southerland 2021; Benoist 2002].

The document's overall framing — each component has precedent but the integrated system (δ-engineered PENK + NEP knockdown + intranasal non-viral co-delivery) is novel and unvalidated — is accurate.

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13. Reproducibility

All results were computed from the public reference sequences named above. Verification artifacts:

- `core_audit.csv` — per-core position/flank audit
- `engineered_protein_check.txt` — 288-aa reconstruction & internal-K/R check
- `dna_audit.txt` — insert translation, cassette, restriction scan
- `gc_repeat_profile.png` — GC + repeat map (Figure 1)
- `peptide_properties.csv` / `peptide_properties.png` — physicochemistry (Figure 2)
- `sirna_anchor_check.txt` — cross-species anchor mapping

- `selectivity_anchors.csv` — 426-ligand ChEMBL δ/μ selectivity landscape (Fig 3)
- `selectivity_pkpd.png` — selectivity ladder + NEP/APN kinetics + occupancy (Fig 3)
- `boltz_selectivity_scores.csv` / `boltz_selectivity.png` — Boltz-2 co-folding ipTM (Fig 4)
- `mmgbsa_results.csv` / `mmgbsa_selectivity.png` / `mmgbsa_selectivity_report.md` — MD + MM-GBSA (Fig 5)
- `FINAL_eng_DOR.dat`, `FINAL_eng_MOR.dat`, `FINAL_nat_DOR.dat`, `FINAL_nat_MOR.dat` — raw gmx_MMPBSA output
- `references.csv` / `references.bib` — bibliography with DOIs
- `P01210.fasta`, `P01210.gff`, `NM_000902.fasta` — input reference sequences

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14. References

Retrieved and verified via PubMed (please cite PubMed); DOIs included where available.

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